Multilevel Models for Airbnb Ratings

An implementation and extension of Chapter 13 of *Statistical Rethinking*

By Tal Ramot, implemented with Claude AI

1 The problem: a rating is not a measurement

A listing with one review rated 5.0 is not better than a listing with four hundred reviews averaging 4.83. The raw average treats them as equals, and that is wrong, because the two numbers are not measured with the same precision.

The data show this before any model is fitted. Across the 70,708 usable London listings, the *mean* rating barely moves with the number of reviews, while the *spread* collapses:

reviews	listings	mean rating	SD of rating
1–2	13,995	4.57	**0.865**
3–5	10,615	4.66	0.496
6–10	9,959	4.70	0.341
11–50	24,332	4.74	0.239
50+	11,807	4.77	**0.179**

The spread shrinks by a factor of almost five while the mean moves by two tenths of a point. If low-review listings were genuinely more variable in quality, we would expect their mean to move too. It does not. What we are looking at is sampling noise: an average over two reviews scatters because it is an average over two reviews.

This is exactly the situation Chapter 13 is about. The chapter's answer is that each listing should be estimated using *all* the data, not only its own, and that the amount of information a listing borrows from the others should depend on how little evidence it has of its own.

2 A generative story, and why it is the Reed frog model

Suppose each review comes from a customer who either had a positive experience or did not. Let p_i be the probability that an arbitrary customer has a positive experience at listing i, and let the overall rating be the fraction of positive experiences, rescaled from $[0, 1]$ to the star range $[1, 5]$. Then from the observed rating y_i and the review count N_i we can recover a count of positive reviews:

$$p_i^{\text{obs}} = \frac{y_i - 1}{4}, \qquad P_i = \text{round}\big(p_i^{\text{obs}}\, N_i\big) \,.$$

P_i is a count of successes out of N_i trials. That is the tadpole data of Chapter 13, with **listings in place of tanks** and **reviews in place of tadpoles**. Every listing contributes exactly one row, exactly as every tank does.

Two assumptions are being made here and it is worth stating them plainly rather than letting them pass. First, `review_scores_rating` is an average of one-to-five star ratings, not a proportion of positives; treating it as one is a modelling choice. Second, `number_of_reviews` counts all reviews, while the rating is computed only from reviews that carried a score, so N_i slightly overstates the number of trials. Neither changes the structure of the problem, but both would change the third decimal place.

A note on vocabulary, since it is easy to get backwards: P_i is the *outcome*, and Binomial(N_i, p_i) is the *likelihood*. The prior lives on the intercepts α_i, not on P_i.

The analysis below uses 200 listings, 40 drawn from each of the five review count bands above, carrying 5,563 reviews between them with a median of 8 per listing. Stratifying rather than sampling at random is deliberate: a random sample would be dominated by the middle of the distribution and would contain too few one-review listings, which are precisely the ones the model is supposed to do something about.

3 Two models

No pooling. Every listing gets its own intercept, and the intercepts know nothing about each other:

$$P_i \sim \text{Binomial}(N_i, p_i), \qquad \text{logit}(p_i) = \alpha_{\text{listing}[i]}, \qquad \alpha_j \sim \text{Normal}(0, 1.5).$$

Partial pooling. The prior over the intercepts is itself estimated from the data:

$$P_i \sim \text{Binomial}(N_i, p_i), \qquad \text{logit}(p_i) = \alpha_{\text{listing}[i]},$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \sigma), \qquad \bar{\alpha} \sim \text{Normal}(0, 1.5), \qquad \sigma \sim \text{Exponential}(1).$$

The second model has two more parameters than the first. It will turn out to be the less flexible of the two, and that apparent contradiction is the whole lesson.

Model names below follow the book. In the code they are spelled out: `m13.1` is `no_pooling`, `m13.1b` is `no_pooling_flat`, and `m13.2` is `partial`, all defined in `src/03-models.R`.

```
m13.1 <- ulam(
  alist(
    P ~ dbinom( N , p ),
    logit(p) <- a[listing],
    a[listing] ~ dnorm( 0 , 1.5 )
  ), data=dat , chains=4 , log_lik=TRUE )

m13.2 <- ulam(
  alist(
    P ~ dbinom( N , p ),
    logit(p) <- a[listing],
    a[listing] ~ dnorm( a_bar , sigma ),
    a_bar ~ dnorm( 0 , 1.5 ),
    sigma ~ dexp( 1 )
  ), data=dat , chains=4 , log_lik=TRUE )
```

One warning about the fixed prior. Ratings here sit near 4.79, which means $p \approx 0.947$ and $\text{logit}(p) \approx 2.9$. A Normal$(0, 1.5)$ prior places the typical listing almost two standard deviations from its own mean, so it is not a weak prior on this scale; it pulls every listing toward $p = 0.5$. To check that the comparison below is not an artifact of that, a third model `m13.1b` was fitted, identical to `m13.1` but with $\alpha_j \sim \text{Normal}(0, 5)$, which really is flat.

4 Fitting the model: geometry, not arithmetic

The multilevel model is harder to sample than it looks. The difficulty is that σ and the intercepts are entangled: when σ is small, the α_j must be packed tightly around $\bar{\alpha}$, and when σ is large they may spread out. The posterior therefore has the shape of a funnel, narrow at one end and wide at the other, and a step size that suits one end of the funnel is wrong for the other.

The standard remedy, and the one the chapter gives, is to *reparameterize*. Instead of sampling the intercepts directly, sample standardized offsets $z_j \sim \text{Normal}(0, 1)$ and rebuild the intercepts from them:

$$\alpha_j = \bar{\alpha} + z_j\,\sigma.$$

This is the same model. It has the same posterior over α_j. But the parameters the sampler actually moves through, z_j, no longer change shape when σ changes, and the funnel goes away.

```
m13.2nc <- ulam(
  alist(
    P ~ dbinom( N , p ),
    logit(p) <- a_bar + z[listing]*sigma,
    z[listing] ~ dnorm( 0 , 1 ),
    a_bar ~ dnorm( 0 , 1.5 ),
    sigma ~ dexp( 1 ),
    gq> vector[listing]:a <<- a_bar + z*sigma
  ), data=dat , chains=4 , log_lik=TRUE )
```

Both versions were fitted. The results:

	divergent	min E-BFMI	max $\hat{R}$	min n_{eff}
centered	0	**0.216**	1.023	**150**
non-centered	0	**0.736**	1.001	**714**

Neither version produced a single divergent transition, so the warning most often associated with the funnel never fired. This is worth being honest about: the textbook symptom did not appear on this data set. The funnel showed up anyway, in two quieter places. Stan warned that the energy diagnostic E-BFMI was below 0.3 on all four chains of the centered model — it came in at 0.216, against 0.736 after reparameterizing — and the centered model extracted only 150 effectively independent draws of σ from the same 2,000 samples that gave the non-centered model 714. Reparameterizing bought nearly a fivefold improvement in the precision of σ for no extra computation, and since everything in Section 9 depends on the σ posterior, that matters.

What the diagnostics mean

Divergent transitions. Stan simulates a physical trajectory across the posterior. When the surface curves more sharply than the step size can follow, the simulated trajectory diverges from the true one, and Stan records it. A divergence means the sampler could not go where it was trying to go, so that region of the posterior is under-explored. Even a handful is a reason to distrust the result.

E-BFMI looks at the same problem from the direction of energy. If the sampler cannot change energy efficiently between iterations, it cannot move between the narrow and wide parts of a funnel. Values below 0.3 are a warning. It catches funnels that produce no divergences at all, which is what happened here.

$\hat{R}$ compares the variance within each chain to the variance across chains. If the chains have converged to the same distribution, the two agree and $\hat{R}$ approaches 1. Values above about 1.01 mean the chains have not agreed yet.

n_{eff} (`ess_bulk`) is the number of *effectively independent* samples. Markov chain draws are correlated, so 2,000 samples are worth fewer than 2,000 independent ones. It is a measure of how much information the chains actually delivered, not how long they ran.

4.1 Trace plots

A trace plot draws each chain's path over the iterations. What we want is what McElreath calls a fuzzy caterpillar: the chains should sit in the same horizontal band, wiggle rapidly, and none of them should wander off or stick in one place for a stretch.

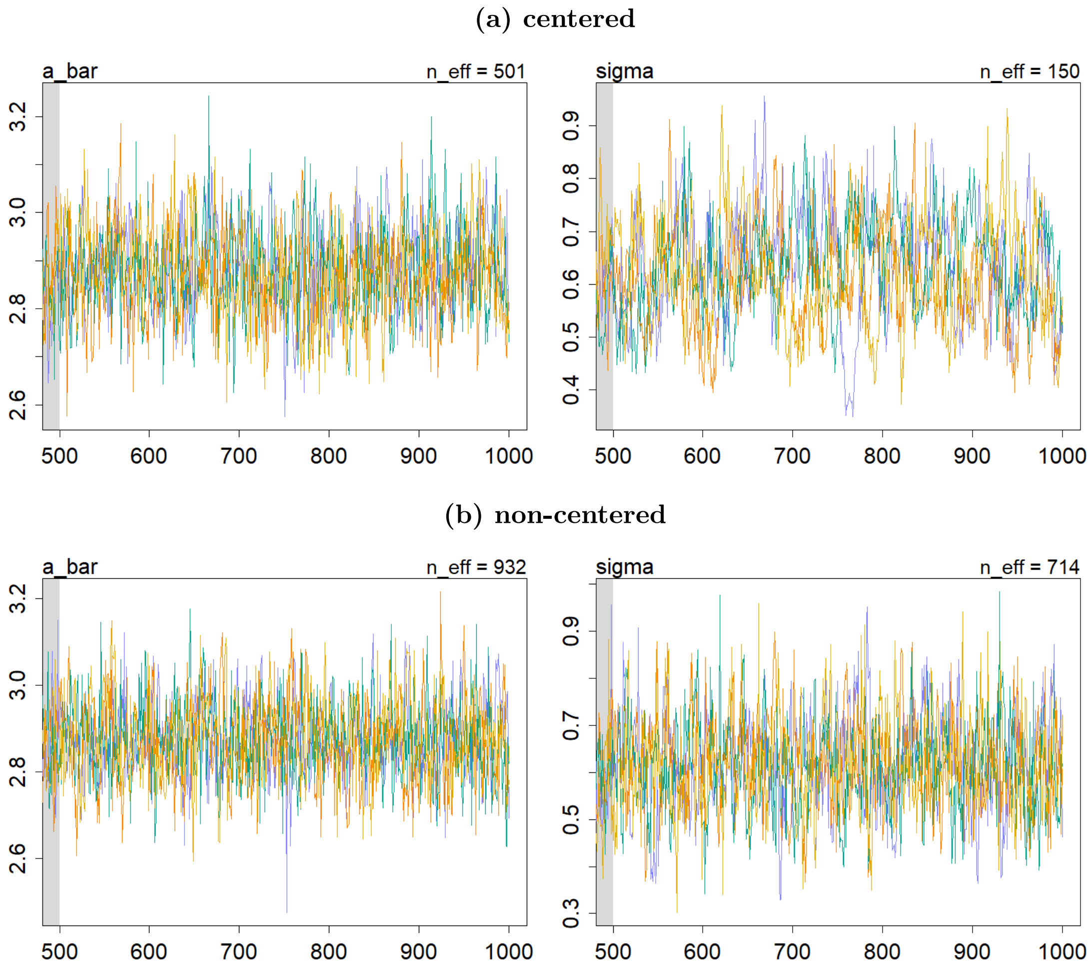


Figure 1: Trace plots for $\bar{\alpha}$ and σ, warmup excluded. Both are stationary — neither is broken — but they are not equally good. The centered σ trace makes slow excursions that last tens of iterations, wandering low for a stretch and then high. The non-centered trace oscillates rapidly with no such drift. Slow excursions mean successive draws carry nearly the same information, which is exactly what a low effective sample size reports.

4.2 Trace rank plots

Trace plots become hard to read once four chains overlap, because the ink piles up and a chain that is slightly wrong is hidden behind three that are right. A trace rank plot fixes this. All the samples from all the chains are ranked together, from lowest to highest, and then each chain's ranks are histogrammed separately.

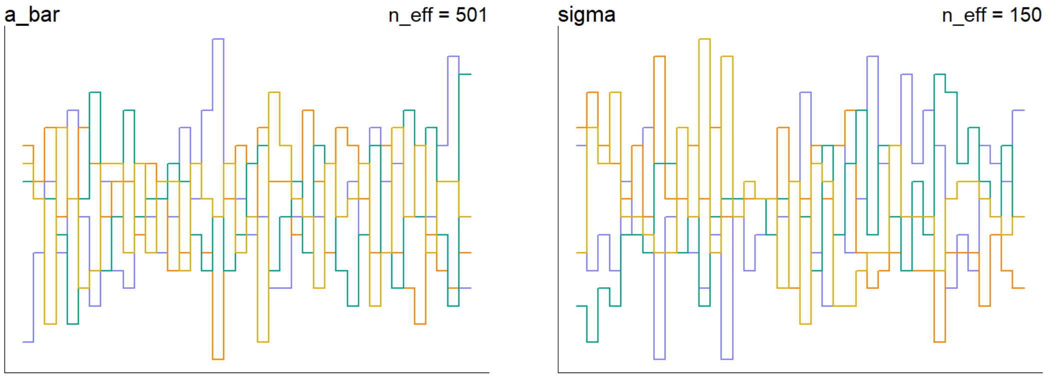
The logic is simple. If every chain is exploring the same distribution, then each chain should own roughly its fair share of the low ranks, the middle ranks, and the high ranks. So the histograms should be flat and should overlap. A chain that is stuck somewhere too long will own too many ranks in one region and show up as a histogram that rises above the others.

5 What the model estimates

Reading this table: `mean` and `sd` describe the marginal posterior of each parameter, and the two percentage columns give the 89% compatibility interval. The last two columns are the diagnostics from the box above, and both are healthy here.

The numbers themselves say two things. $\bar{\alpha} = 2.88$ on the log-odds scale is $\text{logit}^{-1}(2.88) = 0.947$ as a probability, or $1 + 4(0.947) = 4.79$ on the star scale. That is the average listing, and it

(a) centered



(b) non-centered

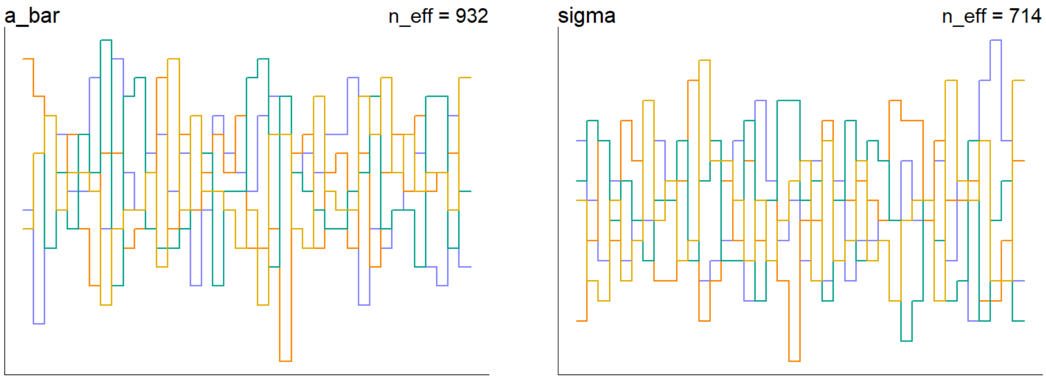


Figure 2: Trace rank plots. Look at the σ panel of (a): one chain holds far more than its share of the lowest ranks at the left edge, and another spikes at the highest ranks on the right. Those are chains that spent too long in one region — the same slow excursions the trace plot showed, now visible as an imbalance rather than as a wiggle. In (b) the four histograms interleave much more evenly.

	mean	sd	5.5%	94.5%	$\hat{R}$	n_{eff}
`a_bar`	2.876	0.094	2.730	3.031	1.001	932
`sigma`	0.617	0.103	0.456	0.786	1.001	714

Table 1: `precis(m13.2, depth=1)`. The 200 listing intercepts are hidden; `depth=2` would show them all.

matches the population mean of the full data set. And $\sigma = 0.62$ says how far listings genuinely differ from one another once sampling noise has been accounted for. It is not zero, so listings really do vary; it is also well below 1.5, which is the fixed prior the no-pooling model was using, and that comparison is what decides the next section.

	WAIC	SE	dWAIC	dSE	**pWAIC**
`m13.2` multilevel	485.1	28.4	0.0	–	**37.5**
`m13.1b` no pooling, flat	508.3	25.7	23.2	14.7	73.3
`m13.1` no pooling, book	690.2	18.3	205.1	18.7	112.1

Table 2: `compare(m13.1, m13.1b, m13.2)`.

6 Comparing the models

WAIC and pWAIC

WAIC estimates how well a model will predict data it has not seen. It takes the fit to the data it *has* seen and subtracts a penalty for flexibility. Lower is better, and only differences between models on the same outcome mean anything — the absolute value does not.

pWAIC is that penalty: the *effective number of parameters*. It is computed from how much the log-likelihood of each observation varies across the posterior. A parameter that is pinned down by the data, or held in place by a tight prior, contributes little. A parameter free to chase its own data point contributes about one. So pWAIC counts not how many parameters a model has, but how many it is really *using*.

dWAIC is the difference from the best model and **dSE** is the standard error of that difference — the number to compare it against.

The pWAIC column is the point of the whole exercise. The multilevel model has 202 parameters. The no-pooling models have 200. Yet the multilevel model uses the equivalent of 37.5 of them, against 73.3 and 112.1. It is the model with the most parameters and the least overfitting.

The reason is the adaptive prior. In `m13.1` each intercept is regularized by a fixed Normal$(0, 1.5)$ that was chosen before seeing any data. In `m13.2` the model learns from the data that the true spread is $\sigma = 0.62$, which is far tighter, and applies that. Regularization the model works out for itself beats regularization we guessed in advance.

The sensitivity fit is worth a sentence, because it comes out backwards from what one might expect. `m13.1b`, with the *wider* prior, has the *lower* pWAIC of the two no-pooling models. The reason is that Normal$(0, 1.5)$ is not merely tight, it is tight *in the wrong place*: it is centered on $p = 0.5$ while the data sit at $p = 0.947$. Prior and likelihood pull against each other for every listing, and that conflict inflates the variance of the log-likelihood, which is what pWAIC measures. The comparison survives either way — 37.5 beats both — so the multilevel model's advantage is not an artifact of the prior we happened to choose.

7 The population of listings

A multilevel model does not only estimate 200 intercepts. It estimates the *distribution* those intercepts were drawn from, and that is what allows it to say anything at all about a listing it has never seen.

```
post <- extract.samples( m13.2 )
plot( NULL , xlim=c(0,6) , ylim=c(0,0.8) ,
     xlab="log-odds of a positive experience" , ylab="density" )
for ( i in 1:100 )
    curve( dnorm(x,post$a_bar[i],post$sigma[i]) , add=TRUE ,
          col=col.alpha("black",0.2) )

sim_listings <- rnorm( 8000 , post$a_bar , post$sigma )
dens( inv_logit(sim_listings) , lwd=2 , adj=0.1 )
```

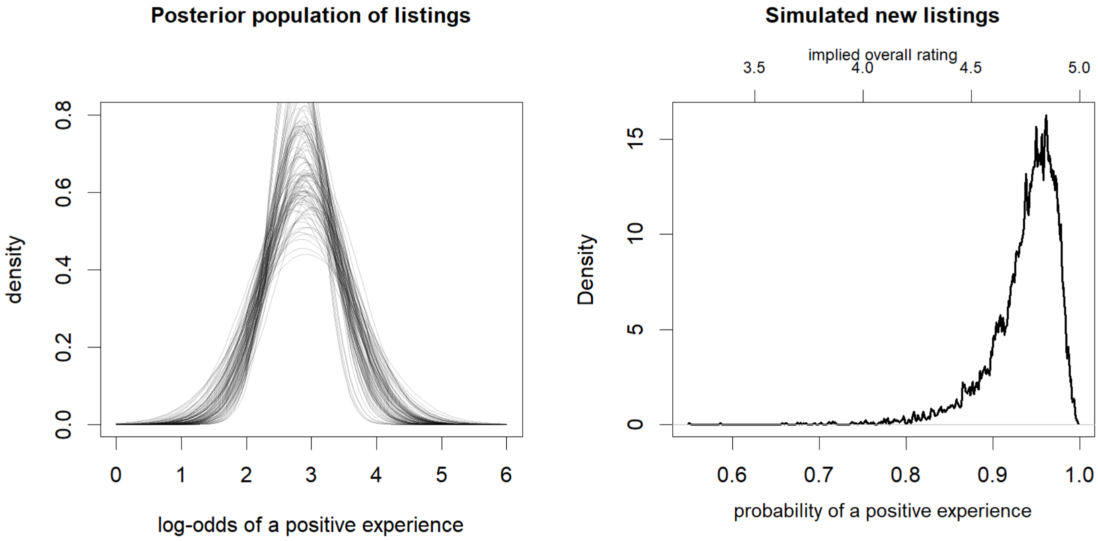


Figure 3: Left: one hundred draws of the population distribution of listings on the log-odds scale, each curve a Normal($\bar{\alpha}, \sigma$) using one posterior sample. The spread of the curves is uncertainty about the population; the width of each curve is the variation among listings. Right: 8,000 simulated listings pushed through the inverse logit, so the horizontal axis is the probability of a positive experience, with the implied star rating marked on top.

The transformation matters. On the log-odds scale the population is a symmetric bell. On the probability scale it is strongly skewed, piled up near 1 with a long tail reaching down. That asymmetry is not something we assumed; it is what a symmetric distribution becomes once it is squeezed through the inverse logit near the top of its range. It is also a fair description of Airbnb: most listings are fine, and the interesting variation is all on the downside.

8 Shrinkage

The picture is the one Chapter 13 promises. On the left, where listings have one or two reviews, the segments are long and every one of them points toward the dashed line. On the right, where listings have fifty reviews or more, the open and filled points sit on top of each other. Nothing was done to the model to produce this gradient. It follows from the binomial likelihood: a count out of 400 trials is sharp evidence and overwhelms the prior, while a count out of one trial is almost no evidence and the prior wins.

reviews	mean shift	spread before	spread after	**shrink factor**
1–2	0.327	0.314	0.032	**0.899**
3–5	0.396	0.445	0.059	0.869
6–10	0.214	0.257	0.045	0.826
11–50	0.111	0.168	0.063	0.625
50+	0.047	0.116	0.070	**0.395**

Table 3: All quantities in rating points. "Spread" is the mean distance from the population mean, before and after pooling; the shrink factor is 1 − after/before.

One column in that table is not monotone and the reason is instructive. The mean shift for one-to-two-review listings (0.327) is *smaller* than for three-to-five (0.396). This is not an error. A listing with a single review is almost always rated exactly 5.0, and 5.0 is only about 0.2 rating

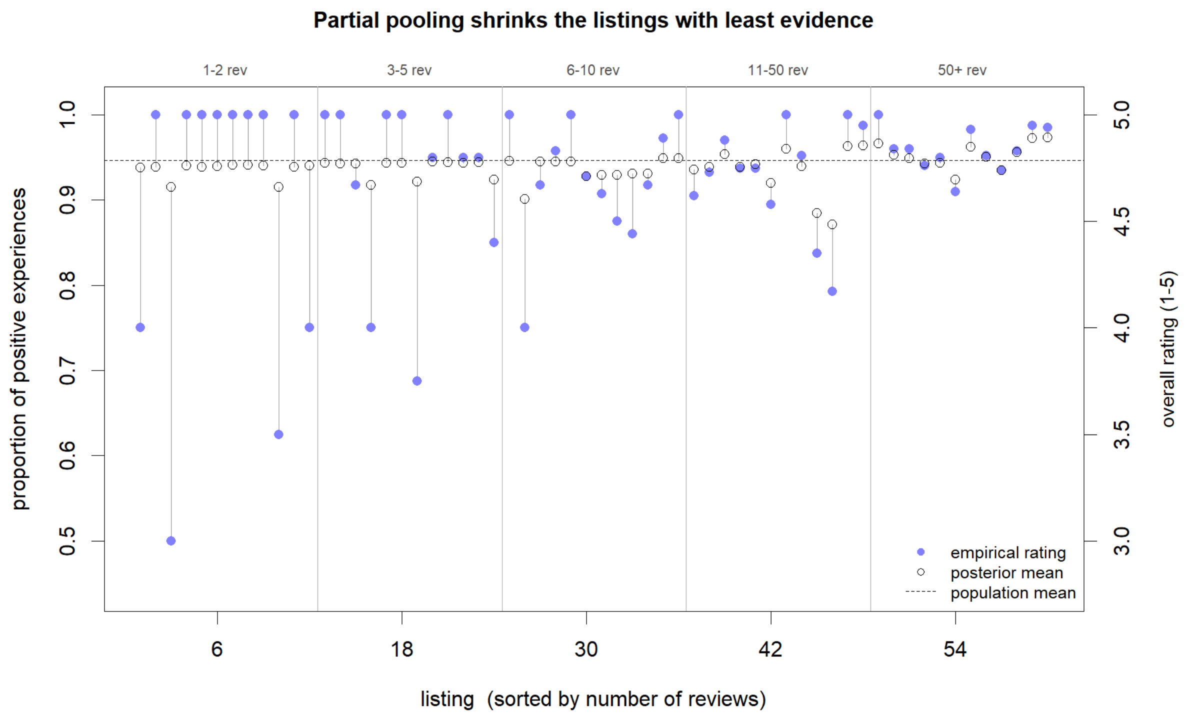


Figure 4: Filled points are the empirical ratings, open points the posterior means, and the segment between them is what pooling did. The dashed line is the population mean $\text{logit}^{-1}(\bar{\alpha})$. Listings are sorted by review count, with the bands marked along the top.

points above the population mean of 4.79, so there is very little distance to travel even though nearly all of it is taken away. The shrink factor, which measures the *fraction* of the gap that pooling closes, is the honest summary, and it falls monotonically from 0.90 to 0.40.

9 Extension: what is σ actually doing?

Everything above rests on a claim that Chapter 13 makes but does not check: that σ is a regularization strength, and that the model chooses a good one. That claim is testable.

If σ really is a regularizer, then fixing it by hand at the value the model chose ought to predict new data better than fixing it anywhere else. So: refit the multilevel model twelve times with σ *pinned* at values from 0.15 to 2.5, leaving $\bar{\alpha}$ free, and score each fit by cross-validation. If the claim holds, the cross-validation curve should peak where the posterior of σ peaked in the model that estimated it.

```
sigma_grid <- exp( seq( log(0.15) , log(2.5) , length.out=12 ) )

for ( i in seq_along(sigma_grid) ) {
    dat_s <- c( dat , list( sigma_fixed=sigma_grid[i] ) )
    fits[[i]] <- ulam(
      alist(
        P ~ dbinom( N , p ),
        logit(p) <- a[listing],
        a[listing] ~ dnorm( a_bar , sigma_fixed ),
        a_bar ~ dnorm( 0 , 1.5 )
      ), data=dat_s , chains=4 , log_lik=TRUE )
}
```

9.1 Cross-validation needs care here

The obvious tool is `PSIS`, and the obvious tool is the wrong one. Every listing contributes exactly one observation, so dropping listing i removes *all* the information about α_i. PSIS estimates what would happen if we dropped a point by reweighting a posterior that was fitted *with* that point, and when the point is the only thing holding a parameter in place, that reweighting is being asked to do far too much. The Pareto k diagnostic says so loudly: essentially every point is flagged.

But the same fact that breaks PSIS makes the exact answer easy. With α_i unconstrained by the remaining data, the leave-one-out prediction for listing i is just the prediction for a brand new listing:

$$p(P_i \mid y_{-i}, \sigma) = \int \text{Binomial}\big(P_i \mid N_i, \text{logit}^{-1}(a)\big) \, \text{Normal}(a \mid \bar{\alpha}, \sigma) \, da,$$

averaged over the posterior of $\bar{\alpha}$. This needs no importance sampling at all — draw $\bar{\alpha}$ from the posterior, draw a around it, average the binomial density. It handles α_i exactly. Its only approximation is that it reuses the full-data posterior of $\bar{\alpha}$ rather than refitting without listing i, which moves $\bar{\alpha}$ by a fraction of its own posterior standard deviation when one listing in two hundred is removed.

9.2 The result

σ	elpd PSIS	bad k	elpd WAIC	**elpd direct**
0.150	−268.0	2	−267.6	−265.7
0.194	−265.2	4	−264.5	−263.1
0.250	−260.6	3	−259.3	−260.0
0.323	−257.1	7	−254.2	−256.8
0.417	−252.3	6	−248.7	−253.9
0.539	−249.4	16	−244.1	**−252.2**
0.696	**−247.0**	27	−240.0	−252.3
0.899	−247.4	36	−238.1	−255.3
1.161	−248.2	45	**−237.2**	−261.8
1.499	−252.4	58	−237.7	−272.2
1.936	−257.7	72	−239.7	−286.6
2.500	−264.4	97	−241.9	−304.4

Table 4: Each row is a separate model with σ pinned. Higher elpd is better; the best value in each column is bold. "bad k" counts the listings, out of 200, whose Pareto k exceeds 0.7 — the diagnostic that says PSIS should not be believed.

The direct cross-validation curve and the posterior land in the same place. The curve is highest at $\sigma = 0.539$, with $\sigma = 0.696$ all but tied (−252.2 against −252.3), and the posterior mode of 0.623 falls between those two adjacent grid points. Both sit comfortably inside the 89% interval $[0.456, 0.786]$.

A purely predictive criterion, which knows nothing about priors, and a purely Bayesian one, which never simulated a held-out data point, choose the same amount of pooling. The multilevel model found by regularization what cross-validation found by brute force.

Three honest caveats. The two quantities are not required to agree *exactly*: they optimize related but different objectives, and the Exponential(1) prior pulls the posterior mode slightly down relative to the cross-validation optimum. The grid is coarse, so it locates the optimum only to the nearest grid point. And the peak is *flat*, so the data do not distinguish sharply between

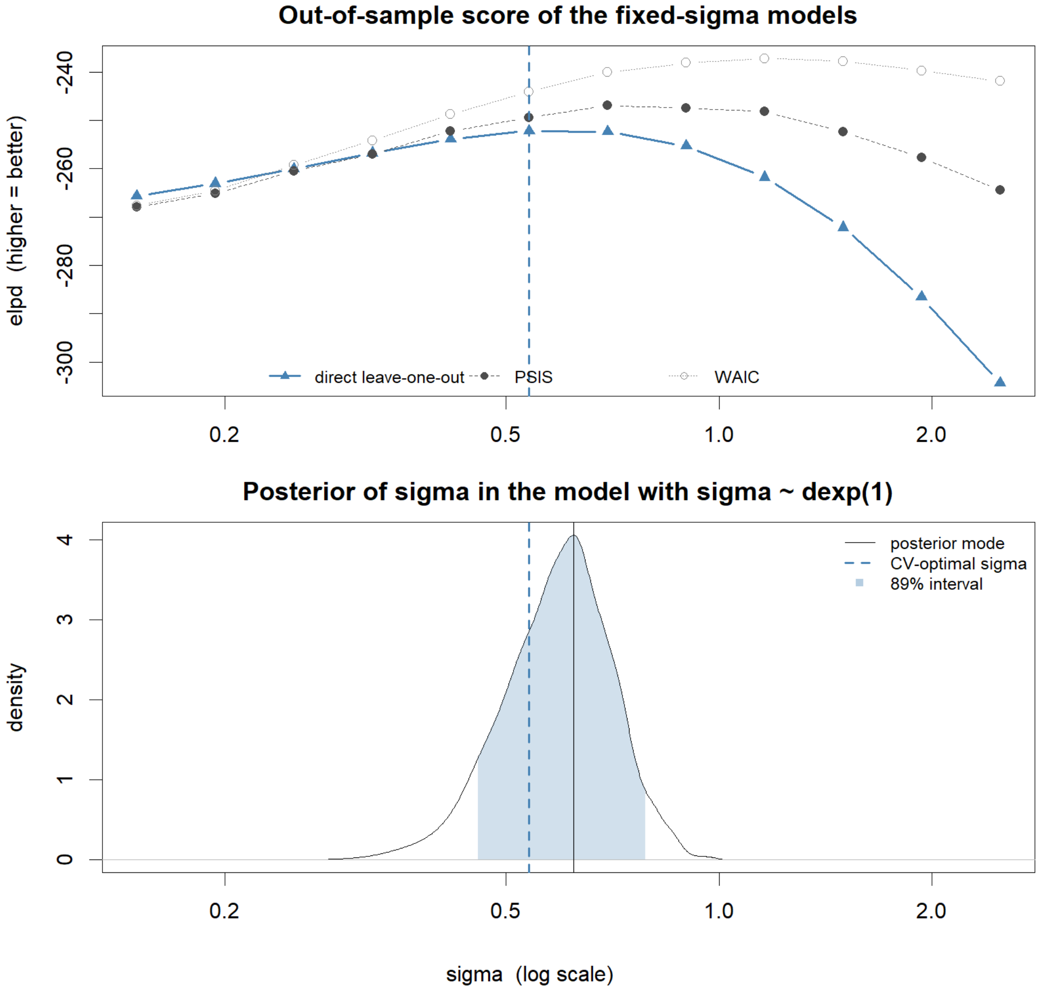


Figure 5: Top: out-of-sample score of each fixed-σ model, by three criteria. Bottom: the posterior of σ from the model that estimated it with a Exponential(1) prior, with its 89% interval shaded. The dashed line is the cross-validation optimum, carried down from the top panel.

nearby amounts of pooling. What the curve does say sharply is that both ends are bad: pinning σ at 0.15 costs 13 elpd and pinning it at 2.5 costs 52. The claim is agreement, not identity.

The disagreement between the criteria is a result in its own right. Both PSIS and WAIC peak at *larger* σ than the direct computation — one grid step for PSIS, three for WAIC — that is, they recommend *less* pooling than is good for prediction. The reason is the same one that made PSIS unreliable: both estimate the cost of a held-out point using a posterior in which that point's own parameter was fitted to it. They therefore undercount how much flexibility the model is using, and a model whose flexibility is undercounted looks like it can afford more.

The bad-k column tracks the damage. It climbs from 2 at $\sigma = 0.15$ to 97 at $\sigma = 2.5$, and stands at 27 at the very σ that PSIS nominates — the diagnostic is loudest exactly where the answer is being read off. The direct computation has no such blind spot. It is a useful reminder that an information criterion is an approximation, and that approximations have assumptions — one observation per cluster violates this one.

10 Conclusion

Starting from the observation that a rating computed from two reviews is not the same kind of number as a rating computed from four hundred, we wrote down a generative story in which reviews are trials and ratings are proportions, and that story turned out to be the Reed frog model. Fitting it required attention to the geometry of the posterior rather than to the model itself: the reparameterization $\alpha_j = \bar{\alpha} + z_j\sigma$ changes nothing about what is being estimated and everything about how well it can be estimated.

The comparison delivered the chapter's central result. The multilevel model has two more parameters than the no-pooling model and uses a third as many of them, because it learned its own regularization instead of being handed one. The shrinkage plot showed where that regularization goes: almost entirely onto the listings with the least evidence, and almost not at all onto the listings with the most.

Finally, pinning σ by hand and scoring the result by cross-validation confirmed that the σ the model estimates is the σ that predicts best. Along the way it also showed that the standard tools for that scoring, PSIS and WAIC, cannot be trusted when every cluster holds a single observation — which is worth knowing, because it is a common shape for data to have.